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ARTICLES

Determination of the Parameters of Molecular-Weight Distribution of Hydroxyethyl Starches by Diffusion-Ordered NMR Spectroscopy

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Abstract—The possibility of the application of diffusion-ordered NMR spectroscopy to the study of the molecular-weight distribution of hydroxyethyl starch is investigated. The use of regression equations relating these parameters to the coefficients of the self-diffusion of polymer macromolecules as a whole, namely, the average coefficient of self-diffusion D_s and the coefficient of self-diffusion at the peak maximum D_p, is proposed for the quantitative assessment of weight-average (M_w) and number-averaged (M_n) molecular weights and molecular weight at the peak maximum (M_p). It is shown that the determination of M_w, M_n, M_p based on the data on D_s gives the best results. It was found that the values of polydispersity indexes found using diffusion-ordered NMR spectroscopy are independent of the choice of diffusion parameter D_s or D_p.

Keywords: diffusion-ordered NMR spectroscopy, parameters of the molecular-weight distribution of polymers, average coefficient of self-diffusion, self-diffusion coefficient at the peak maximum, hydroxyethyl starch

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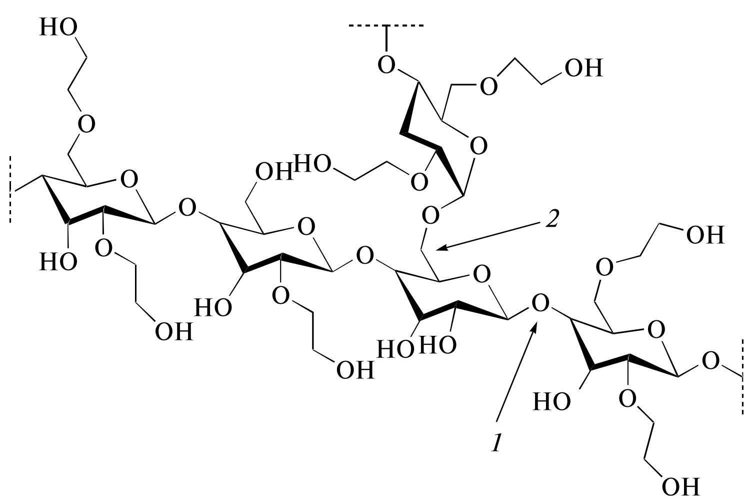
Hydroxyethyl starches (HESs) are efficient plasma volume expanders in infusion therapy; their basic function is the completion of the intra vascular volume of liquid [1, 2]. The molecular structure of HESs represents a polydisperse branched amylopectin, an α-*D*-glucopyranose polymer with substituted hydroxyethyl groups. The linear part of HES macromolecules contains α (1 ⟶ 4) bonds and branches, mainly α (1 ⟶ 6) bonds (see scheme).



A fragment of molecular structure of HESs: (*1*) α (1 ⟶ 4) bond, (*2*) α (1 ⟶ 6) bond [3, 4].

Starches of natural origin cannot be used as plasma volume expanders, because they are rapidly hydrolyzed under the action of amylase. Substituted hydroxyethyl groups in the HES molecule decelerate the hydrolysis and, correspondingly, the metabolic degradation and elimination of preparations from the blood flow [5]. The physical and chemical properties, metabolism, and excretion of HESs depend not only on the properties acquired in the hydroxyethylation of starch, but also on its molecular weight distribution (MWD). On the introduction of a polydisperse colloidal solution of HES into the blood circulatory system, small molecules with the sizes smaller than the value of kidney threshold (45–60 kDa) were quickly excreted, whereas larger molecules remained in the blood circulatory system for a long time, depending on their size and rate of decomposition [1]. Therefore, the determination of MWD is an obligatory task in the pharmacopoeia analysis of the quality of HES medicinal substances [6].

The main parameters of the MWD of polymers are the average molecular weight (MW) and polydispersity index, which characterizes the statistical width of MWD. The average MWs of HES-based preparations present in the market vary in range 70–670 kDa [7]. Depending on the averaging method, we can distinguish the weight-average molecular weight M_w (averaging by the weight of macromolecules in the polymer) and the number-averaged molecular weight M_n (averaging by the number of macromolecules in the polymer). The value of M_w is sensitive to high-molecular fractions and that of M_n, to low-molecular fractions of polydisperse polymers. Therefore, the average MW of polymers is often characterized by molecular weight at the maximum of the chromatographic peak M_p; for narrow-disperse samples with close values of MWs of macromolecules, $M_p = (M_w \times M_n)^{0.5}$ [8]. The M_w/M_n ratio determines the polydispersity index of a polymer, from which one can estimate the molecular weight dispersion of the macromolecules and find the number of branching sites in a branched polymer macromolecule [8]. A decrease in the polydispersity index of HESs improves the pharmacokinetics of the preparation and reduces its effect on the rheological properties of blood after the infusion [9].

Now the safest method of the determination of the average MW and polydispersity index of polymers is provided by the method of multidetector gel permeation chromatography (GPC) [10]. Gel permeation chromatography with a light scattering detector and a concentration detector including a viscometer ensures the determination of molecular masses irrespectively of the method of column calibration. In the last decade, a new promising method has been developed for the determination of the average MW of polymers, i.e., diffusion-ordered NMR spectroscopy (DOSY). It ensures measurements of the self-diffusion of various molecular objects (molecules, macromolecules, molecular complexes, supramolecular systems) using a magnetic field gradient [11]. The DOSY method is based on the registration of the loss of phase coherence among nuclear spins because of the translational motion of molecules in this field [12, 13]. The data on diffusion are used for the quantitative assessment of the sizes of polymer macromolecules and, correspondingly, their molecular weights [14–20]. An advantage of the DOSY method is that, along with the diffusion data, it also provides spectral information (values of chemical shifts, multiplicity of signals), which allows the researcher to solve a whole range of problems of the pharmacopoeia analysis of the quality of HES medicinal substances, namely, of the confirmation and quantitative assessment of molar substitution (ratio of the number of hydroxyethyl groups to the total number of glucose molecules) and the C2/C6 ratio (the ratio of the number of hydroxyethyl groups at the second carbon atom to the number of hydroxyethyl groups at the sixth carbon atom in the glucopyranose cycle) within one experiment.

In the DOSY method, a calibration dependence of the parameter of MWD on the coefficient of self-diffusion D (a quantitative characteristic of the diffusion processes) is determined for a series of objects similar to the studied system in nature and then extrapolated to the value of D of the studied object. In determining the correlation equation, a quite important problem is the correct interpretation of the data of a DOSY experiment, because the determination of the value of D for a polymer as a whole is a nontrivial task even if all instrumental sources of errors are considered and the values of D are corrected. The main difficulty is associated with the fact that not all nuclei in the polymer macromolecule are characterized by single values of D.

The aim of this work consisted in a comparison of different methods for the determination of the coefficient of self-diffusion D of HES macromolecules as whole and in the determination of a functional dependence between the main MWD parameters of HESs and D.

EXPERIMENTAL

The objects of study were standard HES samples (American polymer standards corporation, United States). Their MWD parameters are presented in Table 1. ^{1}H and ^{1}H DOSY spectra were recorded on an Agilent DD2 NMR System 600 NMR-spectrometr with a 5-mm inverse multinuclear sensor equipped with a gradient coil at 300 K. The test samples in 5-mg portions were dissolved in 0.5 mL of D_2O (Cambridge Isotope Laboratories). The solutions obtained were 10-fold diluted with D_2O to concentrations of 1 mg/mL and transferred into Shigemi NMR ampoules with glass plungers for the minimization of the effects of convection and pulse heterogeneity of the magnetic field gradient. The coefficients of self-diffusion were measured using the DBPPSTE sequence (DOSY bipolar pulse pair stimulated echo) [21]. Curves of dif-

Table 1. MWD parameters (kDa) of the studied HES samples

Sample	M_p	M_w	M_n	Sample	M_p	M_w	M_n
HETA 6	5.5	6.0	4.0	HETA 9	8.9	9.4	5.6
HETA 18	16.1	18.3	13.0	HETA 60	52.8	59.7	38.1
HETA 88	80.0	88.0	55.1	HETA 130	121.1	130.6	78.4
HETA 160	153.6	159.2	85.9	HETA 175	176.1	206.5	88.9
HETA 230	201.0	229.0	113.8	HETA 250	219.3	250.4	130.5
HETA 337	374.0	337.6	162.6	HETA 380	330.2	379.8	178.6
HETA 460	396.9	459.4	188.5	HETA 600	468.7	608.1	288.0
HETA 1300	1004.0	1312.0	690.3				

fuse attenuation were obtained at a consecutive 15-step linear increase in the pulse amplitude of magnetic field gradient in the range 1.82–53 G/cm at diffusion times $\Delta = 285$ ms constant for all HESs, gradient pulse duration $\delta_{puls} = 2.0$ ms, and relaxation time $d1 = 5$ s. The mathematical processing of the results was done using the DISCRETE method (discrete sum of exponential decays) [22], within which each curve of diffusion decay of a two-component water–HES mixture was represented by a sum of two exponential components. The correlation equations were derived and statistical parameters (coefficients of determination, actual and table values of the Fisher F-test, Student t-test) were calculated using the MS Excel 2007 software at the significance $P = 0.95$ and sample size $n = 12$.

RESULTS AND DISCUSSION

By self-diffusion is meant the spatial movement of molecules in a medium under a thermodynamic equilibrium due to chaotic thermal movement [23]. Self-diffusion is characterized by the coefficient of self-diffusion D, which is numerically equal to the root-mean-square displacement of a molecule within the diffusion time t_d. In the DOSY method, the data on the value of D for a molecular system are obtained from the analysis of diffusion decay, the dependence of the spin echo signal amplitude on the parameters of magnetic field gradient, described by the Stejskal–Tanner function [24]:

$$I(\delta_{puls}, \Delta) = I_0\exp[-D\gamma^2 g^2 \delta_{puls}{}^2(\Delta - \delta_{puls}/3)],$$

where I and I_0 are spin echo signal amplitudes in the presence and absence of a pulse of magnetic field gradient; Δ is diffusion time, ms; δ_{puls} is the duration of gradient pulse, ms; γ is the nuclear gyromagnetic ratio, Hz T^{-1}; and g is magnetic field gradient, T m^{-1}.

It is known that the dependence of D on the size of molecular objects is described by the Stokes–Einstein equation [25]:

$$D = kT/6\pi\eta R_h,$$

where k is the Boltzmann constant, J K^{-1}; T is absolute temperature, K; η is the viscosity of solution, Pa s; and R_h is hydrodynamic radius, m.

Polymer solutions are usually characterized by several values of D, even for narrow MWDs, when the sample consists of a set of macromolecules similar in lengths and weights. One of the reasons for this is existence in of a number of states corresponding to minima in the potential energy surface of the studied system (for example, different conformations or associates with the solvent) and a chemical exchange between them (spin transition or transition of a group of spins between several states). In the case of slow chemical exchange, each set of signals corresponding to different states will be characterized by a certain value of D. Another reason for the experimentally observed dispersion of D values of nuclei in a polymer molecule are different local mobilities of segments of the polymer chain, independent of the molecular weight of the polymer but specific to each particular system polymer–solvent and depending on the measurement temperature. Within the NMR method, systems of nuclear spins characterized by a single value of the studied parameter are usually selected as phases [23]. The phases thus defined have nothing common with the thermodynamic phases of the state of a substance. Polymer solutions are polyphase systems, because they are characterized by a series of D values. The translation mobility of a polyphase polymer macromolecule as whole is often analyzed using the average self-diffusion coefficient (D_s) [23, 25]:

$$D_s = \Sigma p_i D_i,$$

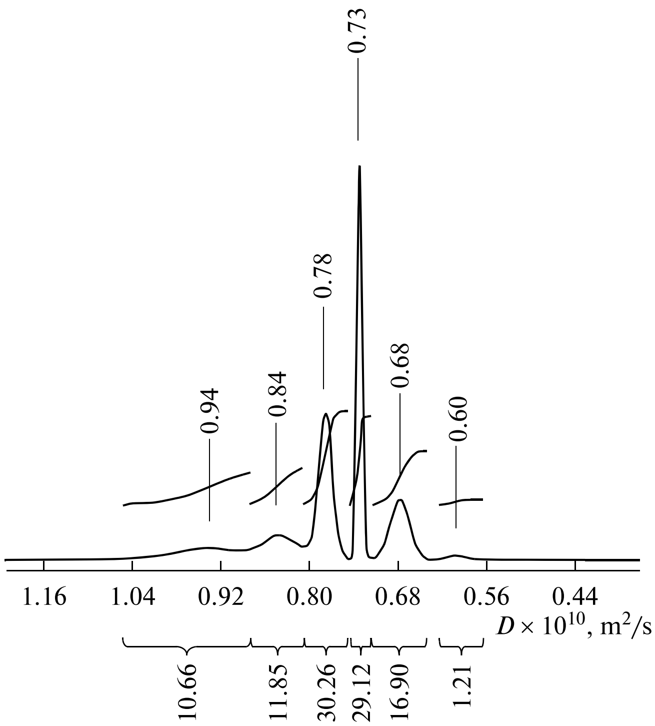


Fig. 1. Distribution of the coefficients of self-diffusion of a HETA 18 HES sample.

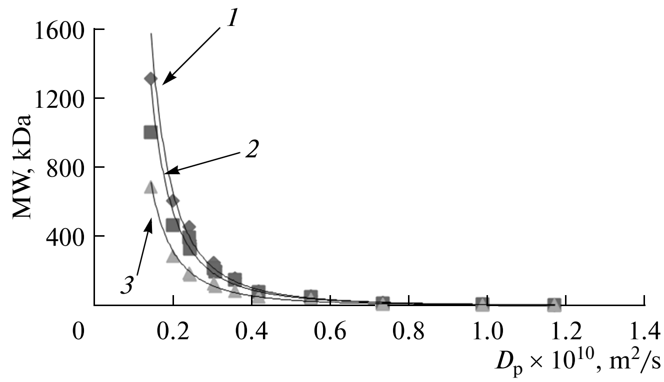


Fig. 2. Average molecular weights (*1*) M_w, (*2*) M_p, and (*3*) M_n of HESs as functions of D_p.

where p_i is the fraction of resonant nuclei in a sample characterized by the value D_i.

In practice, the value of p_i can be determined by integrating peaks in the distribution diagram of D. As an example, Fig. 1 presents a distribution diagram of D for a HETA 18 sample, for which the value of D_s (0.771×10^{-10} m^2/s) was obtained by averaging six values of D_i. The authors of [26–28] proposed to characterize the self-diffusion of a polymer macromolecule as a whole by the value of D in the peak maximum in the distribution diagram of D for a polymer sample (D_p). For example, for a HETA 18 D_p sample, $D_p = 0.731 \times 10^{-10}$ m^2/s (Fig. 1). The D_s and D_p values for HESs with different average MWs found using the parameters of a DOSY experiment and ensuring a high resolution for the values of D of branched polysaccharides (dextrans and HESs) are presented in Table 2 [26, 27]. As can be seen in Table 2, in most cases $D_s > D_p$.

The dependences of the MWD parameters of HESs (M_w, M_n, and M_p) on diffusion parameters (D_p and D_s) were investigated after subdividing the whole data array for the studied HES samples into a training (HETA 6, 9, 18, 60, 88, 160, 230, 250, 380, 460, 600, 1300) and a reference (HETA 130, 175, 337) sample. The regression equations obtained are presented in Table 3 and their graphic views, in Figs. 2 and 3. As can be seen in Table 3, both quantitative characteristics of the processes of self-diffusion (D_p and D_s) correlate with all MWD parameters with a high precision: the values of determination coefficients R^2 in all cases were close to unity. The maximum values of $R^2 = 0.994$ were obtained for regressions I and IV. Therefore, D_p and D_s best correlate with M_w. For the assessment of the importance of coefficients in regression equations, we calculated the values of the Student t-test for each

Table 2. Diffusion parameters of the studied HES samples

Sample	$D_p \times 10^{10}$, m^2/s	$D_s \times 10^{10}$, m^2/s	Sample	$D_p \times 10^{10}$, m^2/s	$D_s \times 10^{10}$, m^2/s
HETA 6K	1.169	1.215	HETA 9K	0.986	1.033
HETA 18K	0.731	0.773	HETA 60K	0.548	0.573
HETA 88K	0.415	0.458	HETA 130K	0.378	0.389
HETA 160K	0.356	0.379	HETA 175K	0.303	0.331
HETA 230K	0.305	0.317	HETA 250K	0.302	0.315
HETA 337K	0.263	0.264	HETA 380K	0.241	0.266
HETA 460K	0.239	0.240	HETA 600K	0.197	0.219
HETA 1300K	0.142	0.145			

Table 3. Regression equations for the dependence of MWD parameters on diffusion parameters of HESs (significance value $P = 0.95$, sample size $n = 12$)

Regression	$\mathrm{MM} = aD^x$	$\log\mathrm{MM} = a\log D + b$	R^2	F	$t(a)$	$t(b)$
I	$M_w = 9.5948\,D_p^{-2.614}$	$\log M_w = -2.6136\log D_p - 25.154$	0.994	1557	−39.46	−36.44
II	$M_p = 9.0314\,D_p^{-2.538}$	$\log M_p = -2.5375\log D_p - 24.419$	0.990	978	−31.28	−28.89
III	$M_n = 6.3558\,D_p^{-2.419}$	$\log M_n = -2.4185\log D_p - 23.382$	0.992	1319	−36.31	−33.69
IV	$M_w = 10.999\,D_s^{-2.620}$	$\log M_w = -2.6201\log D_s - 25.154$	0.994	1517	−38.94	−35.97
V	$M_p = 10.313\,D_s^{-2.544}$	$\log M_p = -2.5438\log D_s - 24.425$	0.990	958	−30.96	−28.59
VI	$M_n = 7.2215\,D_s^{-2.423}$	$\log M_n = -2.4232\log D_s - 23.373$	0.991	1123	−33.51	−31.09

$F_{tab} = 4.96$, $t(a)_{tab} = t(b)_{tab} = 2.22$.

coefficient and Fischer F-tests (Table 3). Calculations of t and F were performed using linear regression equations obtained by the logarithmic conversion of the corresponding power dependences. The plots of linear dependences $\log\mathrm{MW} = a\log D + b$ (IV–VI, Table 3) are shown in Figs. 4 and 5. The calculated values of F and t for the coefficients t_a and t_b were compared with the tabular values. As can be seen in Table 3, the calculated absolute values of F, t_a, and t_b were considerably greater than the table ones, which is indicative of the statistical importance of coefficients in the regression equations.

The obtained regression equations for MWD parameters versus D (I–VI, Table 3) were used for the assessment of M_w, M_p, M_n, and M_w/M_n values for HES samples in the reference sampling (Table 4). The errors of calculations were estimated relative to the experimental values of MWD parameters found by GPC. As can be seen in Table 4, as a whole we observed a good correspondence between the MWD parameters of HESs determined by GPC (Table 1) and those calculated using the proposed regression equations within the DOSY method. Irrespectively of the method of determination of the D value of HESs (D_s or D_p), the correlation between the calculated MWD parameters and the experimental values was improved in the series $M_n \approx M_p < M_w$. The error in the determination of M_w of HESs by the DOSY method virtually

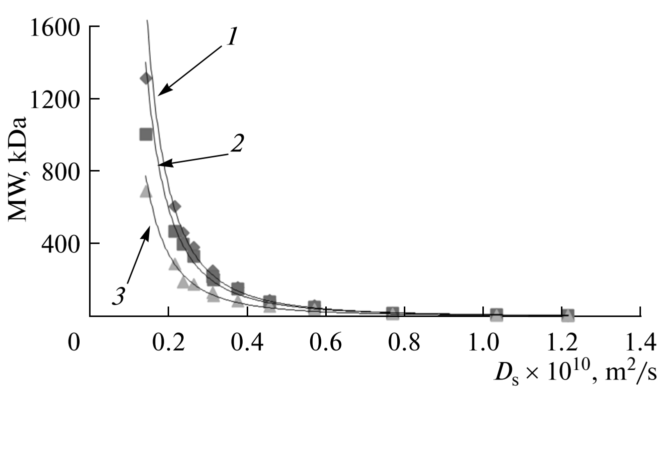


Fig. 3. Average molecular weights (*1*) M_w, (*2*) M_p, and (*3*) M_n of HESs as functions of D_s.

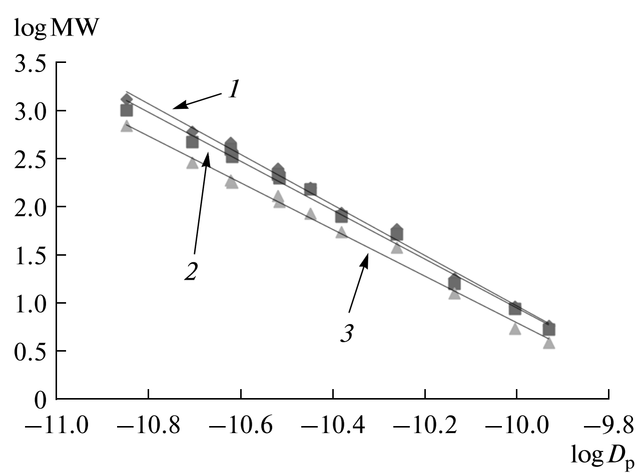


Fig. 4. (*1*) $\log M_w$, (*2*) $\log M_p$, and (*3*) $\log M_n$ as functions of $\log D_p$ for HESs.

Table 4. Calculated values of MWD parameters (kDa) of HES samples of the reference sample

HESs	D	Calculated MWD parameters (X_{DOSY})				δ, rel %*			
		M_w	M_p	M_n	M_w/M_n	M_w	M_p	M_n	M_w/M_n
HETA 130	D_p	122.03	106.68	66.87	1.7	6.56	11.91	14.71	5.88
	D_s	130.52	113.91	71.15	1.7	0.05	5.94	9.25	5.88
HETA 175	D_p	217.54	187.01	114.17	1.9	5.35	6.19	28.43	17.39
	D_s	199.25	171.77	105.22	1.9	3.51	2.46	18.35	17.39
HETA 337	D_p	314.97	267.86	160.80	2.0	6.70	28.38	1.10	4.76
	D_s	360.37	305.37	182.00	2.0	6.75	18.35	11.93	4.76
Average value	D_p					6.20	15.49	14.75	9.34
	D_s					3.44	8.92	13.18	9.34

$^*\ \delta = |(X_{GPC} - X_{DOSY})/X_{GPC}| \times 100\%$.

did not exceed the error of the determination of MW by the GPC method, ~5–10% [29]. A comparison of the average values of relative errors allowed us to draw the following conclusion: the determination of M_n, M_p, and M_w based on the data on D_s led to the best results than the use of D_p values. In spite of the fact that the MW values of different degrees of averaging calculated using D_s were more exact than those calculated using D_p, the polydispersity indexes found from both diffusion parameters coincided and the average relative error of the determination of M_w/M_n did not exceed ~10%. Therefore, in determining the polydispersity index of HESs by DOSY, one can be limited to the data on D_p.

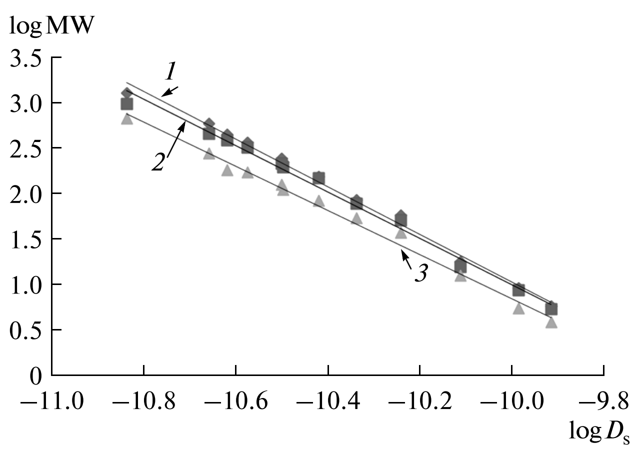


Fig. 5. (*1*) logM_w, (*2*) logM_p, and (*3*) logM_n as functions of logD_s for HESs.

Thus, the DOSY method is suitable for the assessment of the main MWD parameters (M_w, M_n, M_p, polydispersity index) of HESs. In this case, the diffusion parameters most precisely correlate with the weight-average molecular weight of HES. The use of the self-diffusion coefficient D_s of a polyphase HES macromolecule as a quantitative characteristic allowed us to reduce the error of the determination of average molecular weights by the DOSY method to the level of errors of GPC. The work with the coefficient D_p ensures the worse precision of estimates of the average molecular weights of HESs compared to the use of D_s; however, the accuracies of estimates of the polydispersity index were approximately equal.

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